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MILITARY SPECIFICATION

DETECTING SET, MINE, PORTABLE, METALLIC

This limited coordination military specification has been prepared by the Mobility Equipment Research and Development Command, Department of the Army based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-D-23359D. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a portable mine detecting set for locating buried metallic antipersonnel and antitank land mines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling Wood Stringer Construction 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6665

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MILITARY

MIL-B-18	- Battery, Dry.
MIL-I-46058	- Insulating Compound, Electrical (For Coating Printed Circuit Assemblies).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13200E2680	- Detecting Set, Mine Portable, Metallic.
TA13225E8740	- Detecting Set, Mine Portable, Metallic.
13204E8857	- Test Pin.
13220E5627	- Phasing Fixture.
13220E5626	- Magnetite Tube.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The detecting set shall be in accordance with Top Assemblies, TA13225E8740 and TA13200E2680 and as specified herein. The detecting set shall consist of an audio frequency, mutual inductance, phase-selective, bridge-type metallic mine detector; a carrying case; a carrying strap; and a cold weather battery adapter. Unless otherwise specified (see 6.2), two batteries conforming to MIL-B-18, type BA-1568/U (NSN 6135-00-838-0706)

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shall be furnished by the Government for inclusion with each detecting set prior to shipment. The mine detector shall consist of a detecting head, handle, and control box with headset.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more detecting sets for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce detecting sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.4). When specified, the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more detecting sets for inspection as specified in 4.4.

3.4 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-I-46058.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.6 Vibration resistance. The mine detector shall not be damaged by vibration when tested as specified in 4.6.2.3.

3.7 Bounce. The detecting set shall not be damaged when tested as specified in 4.6.2.4.

3.8 Immersion. The mine detector, except headset assembly, shall be watertight and shall not be damaged when submerged under a 5-foot head of water for 24 hours.

3.9 Temperature and humidity.

3.9.1 Operation. The mine detector shall not be damaged when operated under the following conditions.

a. Temperature shocks within the range of plus 50° C to minus 30° C.
b. Ambient temperatures from plus 50° C with solar radiation of 360 British thermal units (Btu) per square foot per hour to minus 30° C without benefit of solar radiation, at all relative humidities possible at these temperatures.

3.9.2 Storage. The detecting set shall not be damaged by storage under the following conditions.

a. Ambient temperatures from plus 70° C to minus 55° C at all relative humidities possible at these temperatures.
b. Elevations from sea level to 40,000 feet and temperatures to minus 55° C as encountered during transportation by air.

3.10 Electrical balance. The mine detector with the detecting head at least 3 feet from metallic objects shall be balanced using the external control. The following output voltages (see 6.3.2), shall be obtained.

a. 15 millivolts alternating current root mean square (mvac rms) or less in high-sensitivity position.
b. 5 mvac rms or less in intermediate-sensitivity position.
c. 1.5 mvac rms or less in low-sensitivity position.

3.10.1 Conditions. The above output voltages shall be obtained with the mine detector operating under each of the following conditions:

a. With the detecting head in air.
b. Over every type of soil including soils having a magnetic volume susceptibility of between 13,000 and 15,000 micro centimeter-gram-second (micro cgs) units.
c. Submersed in fresh or sea water.
d. Throughout the range of operating temperatures of minus 30° C to plus 50° C (ambient).
e. Within a battery voltage range from positive 11.25 volts to positive 14 volts.

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3.10.2 Control position. At 25° C plus or minus 5° C, the output voltage shall be obtained with the balance control position within plus or minus two turns of the center of the range.

3.11 Drift, temperature cycling. The mine detector shall have an output voltage of not more than 50 mvac rms in the intermediate-sensitivity position during four continuous temperature cycles at the rate of 0.5° C per minute from 5° C above to 5° C below the initial ambient temperature.

3.12 Drift, solar radiation, and shade. The mine detector shall have an output voltage of not more than 50 mvac rms in the intermediate-sensitivity position during four continuous cycles of 10 minutes under 360 Btu per square foot per hour of incident solar radiation on the detecting head and 30 minutes in shade.

3.12.1 Chamber. The test chamber volume shall be a minumum of ten times that of the volume of the envelope volume of the test item. The chamber's simulated solar radiation source area shall be a minimum of 125 percent of the horizontal area projection of the test item.

3.12.2 Solar radiation source. For the purposes of this test, the following spectral distribution of solar radiation is acceptable: 50 to 72 watts/ft^2 of infrared (of wavelengths above 7,800 angstrom units), 4 to 7 watts/ft^2 of ultraviolet (of wavelengths below 3,800 angstrom units), and the balance visible. The radiation source shall be located at least 30 inches away from any outer surface of the test item. (Lamp vendor's spectral distribution curves may be used in establishing the spectral distribution within the above specified limits. U.S. Bureau of Standards traceability of this vendor data is waived).

3.13 Detection capabilities. The mine detector shall detect a copper test pin conforming to drawing 13204E8857 when the pin is positioned 3 inches from the detecting head with a nonmetallic medium or any combination of nonmetallic media (see 6.3.1), between the pin and the detecting head. Detection of the copper pin shall be indicated by an increase in output voltage under each of the following sensitivity switch control positions.

a. High sensitivity. An increase of 325 mvac rms plus or minus 120 mvac rms above any balance level of less than 15 mvac rms.
b. Intermediate sensitivity. An increase of 75 mvac rms plus or minus 28 mvac rms above any balance level of less than 5 mvac rms.
c. Low sensitivity. An increase of 19 mvac rms plus or minus 7 mvac rms above any balance level of less than 1.5 mvac rms.

3.14 Static magnetic rejection. The mine detector shall have an output voltage of 100 plus or minus 10 mvac rms when each side of the detecting head is exposed to the magnetite tube, drawing 13220E5626. This output voltage shall be balanced for the left side and for the right side to within 10 mvac rms.

3.15 Dynamic magnetic rejection. The detecting head shall be swept at a constant rate of 2 feet per second at a height of 3 inches above a lane

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containing 4-inch by 4-inch wood ties 4 feet long, buried in soil having a magnetic volume susceptibility of between 13,000 and 15,000 micro cgs units or an equivalent field simulator. The mean output voltage response obtained from the wood ties shall not be greater than 1.5 times the output voltage response obtained from the copper test pin (drawing 13204E8857) positioned flush with the lane, when tested in accordance with 4.6.2.12.

3.16 Overload recovery time. A 6-inch diameter by 1/8-inch steel disk shall create a saturating signal. The mine detector shall return to electrical balance (see 3.10) within 0.5 seconds after removal of the disk from the detecting head.

3.17 Detecting-head shielding. The detecting head shall give a minimum response of 20 decibels (dB) for each receiver coil and a minimum response of 20 dB for the transmitter coil.

3.18 Current drain. The mine detector shall have a current drain of not more than 20 milliamperes direct durrent (dc) throughout the battery voltage range specified in 3.10.1.e.

3.19 Reliability. The mine detector shall demonstrate a specified Mean-Time-Between-Failure (MTBF) of 1000 hours and a lower test MTBF shall be 500 hours when tested in accordance with 4.6.2.15.

3.20 Low voltage aural monitor. The mine detector at 25° C plus or minus 5° C shall indicate when the battery voltage has decreased to 10.70 volts plus 0.4, minus 0.3 volts dc. Once the alarm actuates, the signal in the headset shall remain on until the battery voltage drops to 7.5 volts dc or less. The amplitude of the alarm signal shall be greater than the detection signal (see 3.13), at the point of actuation of the low voltage alarm. The fundamental frequency of the alarm shall be between 100 and 200 Hertz (Hz).

3.21 Drift, null. The mine detector output voltage shall not change more than 15 mvac rms in the high sensitivity position under an ambient temperature of 25° C plus or minus 5° C during a 30-minute period.

3.22 Identification marking. The detecting set shall be identified where specified herein in accordance with MIL-STD-130. The marking shall be applied to plates as specified on the drawings. The plates shall be attached as specified on the drawings.

3.23 Treatment and painting. The portions of the detecting set normally painted shall be cleaned, treated, and painted as specified on the drawings.

3.24 Workmanship. The workmanship in the detecting set shall be in accordance with Requirement 9 of MIL-STD-454.

3.25 Safety. The equipment shall be designed to eliminate unnecessary sharp edges, excessive length of fastening devices and shall meet applicable criteria of MIL-STD-454, Requirement 1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.3.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The detecting set shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The mine detector shall be subjected to the tests indicated by "X" in column 1 of table II. Failure of any test shall be cause for performing the inspection specified in 4.1.1.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production detecting sets will be selected at random by the Government

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from the detecting sets being produced by the production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 through 4.6.2.17 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production detecting set shall not exclude the remaining detecting sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure Failure of an initial production detecting set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production detecting set(s) and shall be cause for refusal by the Government to continue acceptance of production detecting sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on detecting sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all detecting sets accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examinations and tests shall be in accordance with MIL-STD-105. Lots shall be formulated from detecting sets which have successfully completed the individual tests specified in 4.5.3.1.

4.5.2 Examination. Each sample detecting set selected in accordance with 4.5.1 shall be examined for the major and minor characteristics as specified in 4.6.1 for possible defects. AQL shall be 1.0 percent defective for major defects and 2.5 pecent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each detecting set shall be subjected to the tests indicated by "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests indicated by "X" in column 3 of table II. No adjustment shall be made on the detecting set after completion of the individual tests. AQL shall be 1.0 percent defective.

4.6 Inspection procedure.

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4.6.1 Examination. Each sample detecting set shall be examined for the major and minor characteristics listed in table I for possible defects as specified in 4.5.2.

TABLE I. Examination.

Number	Characteristic	Requirement Paragraph
Major		
101.	Dimensions that affect form, fit, or function.	3.1.1
102.	Materials not as specified.	3.4
103.	Fungus and moisture resistance not as specified.	3.5
104.	Identification marking not as specified.	3.9
105.	Safety requirements not as specified.	3.10
Minor		
201.	Dimensions that do not affect form, fit, or function.	3.1.1
202.	Treatment and painting not as specified.	3.23
203.	Workmanship not as specified.	3.24

4.6.2 Tests. Tests shall be in accordance with table II. A test lane meeting the requirements of 3.15 and 4.6.2.12 will be available to the contractor (see 6.5). Where "damage" is referred to herein, it shall include, but not be limited to, the following:

a. Rough handling - breakage, deformation, loosening, or any condition resulting in malfunctioning of the detecting set.
b. Temperature and humidity - deterioration or corrosion of any hardware or finish, breakage, deformation, or any condition which may result in malfunctioning of the detecting set.

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TABLE II. Tests.

Initial Production & Preproduction	Quality Conformance Ind.	Quality Conformance Samp.	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	-	X	Detecting-head shielding.	4.6.2.1	3.17
X	X	-	Electrical balance.	4.6.2.2	3.10
X	X	-	Detection capabilities.	4.6.2.10	3.13
X	X	-	Static magnetic rejection.	4.6.2.11	3.14
X	-	-	Overload recovery time.	4.6.2.13	3.16
X	-	X	Current drain.	4.6.2.14	3.18
X	X	-	Low voltage aural monitor.	4.6.2.16	3.20
X	-	-	High temperature and humidity storage.	4.6.2.6	3.9
X	-	-	Low temperature and altitude, storage.	4.6.2.7	3.9
X	-	X	Immersion.	4.6.2.5	3.8
X	-	X	Temperature operation and thermal shock.	4.6.2.8	3.9
X	-	-	Drift, temperature cycling.	4.6.2.9.a	3.11
X	-	X	Drift, solar radiation and shade.	4.6.2.9.b	3.12
X	-	X	Drift, null.	4.6.2.17	3.21
X	-	X	Vibration resistance.	4.6.2.3	3.6
X	-	-	Bounce.	4.6.2.4	3.7
X	-	X	Dynamic magnetic rejection.	4.6.2.12	3.15
X	-	-	Reliability	4.6.2.15	3.19
X	X	X	Safety	4.6.2.18	3.25

4.6.2.1 Detecting-head shielding.

4.6.2.1.1 Test conditions. Conduct tests in a location where stray magnetic and electrostatic fields induce an open-circuit voltage less than 10 mvac rms. Tests shall be performed on detecting heads separate from a complete detector and before cables are attached.

4.6.2.1.2 Equipment. The following equipment shall be used:

a. Test jig in accordance with figure 1.
b. Audio oscillator capable of supplying 10-vrms, 2500-Hz sine wave into a 600-ohm load with less than 5 percent distortion.

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c. High impedance, true rms, ac voltmeter.
d. Shielded lead, including connectors for use from voltmeter to test jig.
e. Shielded lead, terminating with alligator or similar clips for use from audio oscillator to test jig.
f. Hookup wire harness and connectors used to connect all four brass disks on test jig for transmitter coil shield test.

4.6.2.1.3 Procedure.

a. Place the detecting head, terminals up, on the test fixture brass plate. The head should be equidistant from the edges of the plate.
b. Place the test fixture on the head, making sure that the plastic disks on the test fixture are correctly seated in the receiver coil recesses.
c. Connect the voltmeter and oscillator ground leads to the brass plate as shown in figure 2. Connect head terminal No. 5 to the brass plate with a jumper wire.
d. Connect the center conductor of the oscillator shielded lead to head terminal No. 6. Connect the center conductor of the voltmeter shielded lead to disk No. 1. Connect a jumper from the shield of the oscillator lead at brass plate to head terminal No. 2.
e. Adjust the oscillator for 2500 Hz and minimum output voltage. Measure and record the background stray field reading on the voltmeter. Adjust the oscillator for an output increase of 10 to 15 dB above the background stray field pickup as indicated by the voltmeter. Record the reading.
f. Remove the alligator clip from head terminal No. 2, making sure it does not make contact with any conductors, terminals, or brass disks. Record the voltmeter reading. Reattach the alligator clip to head terminal No. 2.
g. Repeat steps e and f with the voltmeter attached to brass disks 2, 3, and 4. This completes the receiver coil toroid test.
h. Connect brass disks 1, 2, 3, and 4 together and attach the center conductor of the voltmeter lead to the connecting harness. Remove jumper wires from head terminal No. 2 and No. 5 to brass plate. Connect head terminals No. 2 and No. 5 together with jumper wire. Remove oscillator center conductor from head terminal No. 6 and attach it to head terminal No. 3. Connect a jumper from head terminal No. 2 to the shield of the oscillator lead (at the brass plate). Repeat step e. Repeat step f, removing the jumper to the oscillator shield from head terminal No. 2. Record the voltmeter reading. This completes the transmitter coil toroid test.
i. Readings from steps e and f should be not less than 20 dB for the receiver coils and not less than 20 dB for the transmitter coil when calculated in the formula:

$$dB = 20 \log \frac{Rf}{Re},$$

where Re = reading of step e, and
Rf = reading of step f, above.

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j. Due to variations in heads, it may be necessary to vary ranges on voltmeter. This does not affect validity of test as long as readings of steps e and f do not fall lower than 10 dB above background stray field pickup.

A response of less than 20 dB from any coil, receiver, or transmitter shall constitute failure of this test.

4.6.2.2 Electrical balance. Position the detecting head not less than 3 feet from metallic objects in an ambient temperature of 25° C plus or minus 5° C. Record the output voltage (see 6.3.2), with a power supply voltage of positive 11.50 volts dc plus or minus 0.25 volts dc under the following conditions:

a. Sensitivity switch in high position.
b. Sensitivity switch in intermediate position.
c. Sensitivity switch in low position.

Repeat the above with a power supply voltage of positive 13.75 volts dc plus or minus 0.25 volts dc. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.2.1 Control positions. Balance the mine detector using the external control to an output voltage of 5 mvac rms or less under the following conditions:

a. Sensitivity switch in the intermediate position.
b. Ambient temperature of 25° C plus or minus 5° C.
c. Detecting head not less than 3 feet from metallic objects.
d. Battery voltage of positive 13.00 volts dc plus or minus 0.25 volt.

Record the positions of the external control. An output voltage of more than 5 mvac rms or the external control positioned more than two turns from the center of the range shall constitute failure of this test.

4.6.2.3 Vibration resistance. Test the mine detector, securely fastened to the vibration table to prevent slippage, in accordance with MIL-STD-202, method 201. The mine detector shall have a battery installed and the sensitivity switch in the off position for the duration of the vibration test. After vibration, examine the mine detector for damage; test in accordance with 4.6.2.2.1. Any damage to the mine detector or failure of 4.6.2.2.1 shall constitute failure of this test.

4.6.2.4 Bounce. Test the detecting set on a package-testing machine capable of maintaining a constant speed of 260 rpm (4-1/3 cycles per second) with an amplitude of 1/2 inch (1 inch total excursion). Place the detecting set between the "fence" with 1-inch spare or "throw." Bounce the detecting set in the three mutually perpendicular directions; that is, the ends, the sides, and the top and bottom of the detecting set case shall be subjected to the bumping, each for a period of 15 minutes (total time 45 minutes). Examine the detecting set for damage; test in accordance with 4.6.2.2.1. Any damage to the detecting set or failure of 4.6.2.2.1 shall constitute failure of this test.

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4.6.2.5 Immersion. Immerse the mine detector, except headset, in water to a depth of 5 feet for a period of 24 hours. The attached short handle shall be positioned so that the straight edge is parallel to the top surface of the detecting head. Visually inspect the mine detector for entrance of water within 30 minutes after completion of the test. Examine the mine detector for damage; test in accordance with 4.6.2.2.1. Failure of 4.6.2.2.1 or nonconformance to 3.8 shall constitute failure of this test.

4.6.2.6 High-temperature and humidity storage. Test the detecting set in accordance with MIL-STD-202, method 106, except the upper temperature limit shall be plus 70° C plus or minus 2° C and step 7 shall not be performed. Polarization and loading voltages are not applicable. Upon completion of step 6 (see MIL-STD-202, method 106) of the final cycle, the detecting set shall be removed from the chamber to an ambient temperature of 25° C plus or minus 5° C. Remove the mine detector from its carrying case, examine for damage, and test in accordance with 4.6.2.2.1. Any damage to the mine detector or failure of 4.6.2.2.1 shall constitute failure of this test.

4.6.2.7 Low temperature and altitude storage. Place the detecting set in the altitude test chamber. Reduce the internal absolute pressure to 5.5 inches of mercury (corresponding to an altitude of 40,000 feet above sea level) and reduce the temperature to minus 55° C plus or minus 2° C. Maintain this pressure and temperature for 72 hours. The pressure in the chamber shall then be increased at a rate of not less than 3.0 inches of mercury per minute until atmospheric pressure has been reached. Remove the mine detector from its carrying case and at least 2 hours later, examine for damage and test in accordance with 4.6.2.2.1. Any damage to the mine detector or failure of 4.6.2.2.1 shall constitute failure of this test.

4.6.2.8 Temperature operation and thermal shock. Place the mine detector in the temperature chamber and reduce the temperature to minus 30° C plus or minus 2° C. Maintain this temperature for 3 hours, then remove the mine detector from the chamber. Test in accordance with 4.6.2.10.2, starting within 10 minutes and completing within 30 minutes after removal from the chamber, using the cold weather battery adapter assembly from the carrying case, and the sensitivity switch in the intermediate position. This test shall be duplicated for six continuous test cycles. Failure of 4.6.2.10.2 shall constitute failure of the low temperature operation test. Place the mine detector (removed from its carrying case and without the cold weather battery adapter assembly) in the chamber and increase the temperature to 50° C plus or minus 2° C. Maintain this temperature for 3 hours, then remove the mine detector from the chamber. Test in accordance with 4.6.2.10.2, starting within 10 minutes and completing within 30 minutes after removal from the chamber. This test shall be duplicated for six continuous test cycles. Failure of 4.6.2.10.2 shall constitute failure of the high-temperature operation test. For conformance sample testing, under this paragraph, only one test cycle is required.

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4.6.2.9 Drift temperature cycling, solar radiation and shade.

a. Place the mine detector in a temperature chamber (see 3.12.1) with the detecting head not less than 3 feet from metallic objects at an ambient temperature of 25° C plus or minus 5° C. With the sensitivity switch in the intermediate position, balance the mine detector to an output voltage of 5 mvac rms or less. Change the chamber temperature at a rate not to exceed 0.5° C per minute first to 5° C below ambient, then to 5° C above ambient and then back to ambient temperature. Subject the mine detector to four continuous cycles as specified above within a time of 48 minutes plus or minus 8 minutes per cycle. Measure the output voltage. An output voltage of more than 50 mvac rms shall constitute failure of the temperature cycling drift test.
b. Solar radiation and shade. Place the mine detector in a chamber with the detecting head not less than 3 feet from metallic objects at an ambient temperature of 25° C plus or minus 5° C. With the sensitivity switch in the intermediate position, balance the mine detector to an output voltage of 5 mvac rms or less. Using lamps (see 3.12.2), whose radiation approximates the spectral characteristics of sun radiation, apply radiation of not less than 360 Btu per square foot per hour to the top of the detecting head for a period of 10 minutes. Turn off the lamps for a period of 30 minutes. Subject the mine detector to four continuous cycles as specified above. Measure the output voltage at the end of each 30-minute off period. An output voltage of more than 50 mvac rms shall constitute failure of this solar radiation and shade drift test. After successful completion of the above tests, remove the mine detector from the chamber and test in accordance with 4.6.2.10.2, except for the test using the cold weather battery adapter assembly. Failure of 4.6.2.10.2 shall constitute failure of this test.

4.6.2.10 Detection capabilities.

4.6.2.10.1 Preproduction model. Balance the mine detector in air with the detecting head not less than 3 feet from metallic objects. Place in operation under the following conditions:

a. The detecting head 3 inches above soil having a magnetic volume susceptibility of between 13,000 and 15,000 micro cgs units with the copper test pin shown on drawing 13204E8857 buried in the vertical position so that its top surface is flush with ground level. The mine detector may be rebalanced after moving the detecting head into the area of the magnetic soil. An equivalent field simulator may be substituted for the magnetic soil and copper test pin. Place the sensitivity switch in the intermediate position. Record the maximum output voltage.
b. The detecting head 2 feet under water and 3 inches above the copper test pin buried flush with the water-ground interface. Place the sensitivity switch in the high position. Record the maximum output voltage.

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c. In air with the copper test pin 3 inches from the top and in the center of each of two electrically opposed receiver coils. Record the output voltage for each of the receiver coils and for each of the sensitivity switch positions.

Nonconformance to 3.13 shall constitute failure of this test.

4.6.2.10.2 Individual. Balance the mine detector in air with the detecting head not less than 3 feet from metallic objects. Place the copper test pin jig in the recess of each of two electrically opposed receiver coils. Record the output voltage for each of these receiver coils for each of the sensitivity switch positions. Repeat the test, in the intermediate sensitivity position only, using the cold weather battery adapter and the replacement battery. Nonconformance to 3.13 shall constitute failure of this test.

4.6.2.11 Static magnetic rejection. Place the detecting head transmitter coil into the hole on the top of the phasing fixture shown on drawing 13220E5627. Balance the mine detector in the intermediate sensitivity position to a minimum output voltage not to exceed 5 mvac rms. Place the magnetite tube shown on drawing 13220E5626 into the slots on the right-hand side of the phasing fixture and, with no adjustment to the detector, measure and record the output voltage. Repeat the measurement with the magnetite tube in the slots on the left-hand side of the phasing fixture. An equivalent test fixture may be used for this test. Nonconformance to 3.14 shall constitute failure of this test.

4.6.2.12 Dynamic magnetic rejection. Mount the mine detector on a fixture and balance to a minimum output voltage not to exceed 5 mvac rms when the power sensitivity selector switch is in the intermediate sensitivity position. Sweep the detecting head at a constant rate of 2 feet per second at a height of 3 inches above a test lane. This lane shall be 30 feet long by 4 feet wide by 3 feet deep and shall be filled with soil having a magnetic volume susceptibility of between 13,000 and 15,000 micro cgs units. The lane shall contain five wood ties 4 inches by 4 inches by 4 feet long, spaced 5 feet apart and oriented perpendicular to the direction of motion, buried so that the top surface is flush with ground level. Record continuously the output voltage during the sweeping operation. An average of the rms (root-mean-square) output voltage responses obtained from the wood ties of more than one and a half times the rms output voltage obtained from the copper pin (see 4.6.2.10.1.a) shall constitute failure of this test. An equivalent field simulator may be substituted for the test lane.

4.6.2.13 Overload recovery time. Position the mine detector in a ambient temperature of 25° C plus or minus 5° C and no less than 3 feet from metallic objects. Place the sensitivity switch in the high position and balance the detector to an output voltage less than 15 mvac rms. Place a 6-inch diameter by 1/8-inch steel disk to within 3 inches of the top side of the detecting head for not less than 5 seconds. Remove the steel disk from the detecting head. Record the time required for the output voltage to traverse from a saturated state to balance (less than 15 mvac rms) on an oscilloscope or oscillograph. Inability of the mine detector to return to balance within 0.5 seconds shall constitute failure of this test.

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4.6.2.14 Current drain. With the sensitivity switch in the high position, place the mine detector in operation in an ambient temperature of 25° C plus or minus 5° C. Use a power supply voltage of positive 13.75 volts dc plus or minus 0.25 volt dc. Measure and record the current drain. A current drain of more than 20 milliamperes shall constitute failure of this test.

4.6.2.15 Reliability. Turn the mine detector on and leave it on a sufficient length of time to determine acceptance or rejection in accordance with MIL-STD-781, Test Plan III C, using the lower test MTBF. Each equipment tested on the reliability test shall operate at least one-half the average of all equipments on the reliabiltiy test. An external power supply may be used. Test the mine detector after each 24 hours of operation in accordance with 4.6.2.2.1 and 4.6.2.10.2 and at completion of reliability test. For the purpose of determining the MTBF, a failure is defined as any malfunction, degradation of performance outside specified limits, or personnel and equipment safety hazard. Normal replacement of batteries is allowable. Nonconformance to 3.19 or inability to pass the tests in 4.6.2.2.1 and 4.6.2.10.2 shall constitute failure of this test.

4.6.2.16 Low voltage aural monitor. Place the mine detector in an ambient temperature of 25° C plus or minus 5° C. Adjust the power source voltage to 11.25 volts dc, balance the detector in air and perform the following:

a. Reduce the power supply voltage until the aural alarm is actuated (a signal in the head set). Measure the power source voltage and output voltage (see 6.3.2).
b. Lower the power supply voltage to 7.5 volts dc ± 1.0 volts dc. Nonconformance to 3.20 shall constitute failure of this test.

4.6.2.17 Drift, null. Place the mine detector in an ambient temperature of 25° C plus or minus 5° C and no less than 3 feet from metallic objects. Place the sensitivity switch in the high position and balance the detector to an output voltage less than 15 mvac rms. Remeasure this output voltage after 30 minutes. A voltage change of greater than 15 mvac shall constitute failure of this test.

4.6.2.18 Safety. The mine detector shall be examined initially and observed throughout testing to determine compliance with 3.25. Nonconformance shall constitute failure of the safety test.

4.7 Inspection comparison. The Government may select detecting sets at any time during the contract production period and subject these detecting sets to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production detecting sets. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison detecting set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparision detecting set(s) and shall be cause for refusal by the Government to continue acceptance of production detection sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on detecting sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all detecting sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

Major

104. Case not closed, secured, and tamperproof seals attached as specified.
105. Box not as specifed for level A.
106. Pallet not as specified for level A.
107. Load arrangement not as specified for level A.
108. Height greater than specified for level A.
109. Strapping not as specified for level A.
110. Battens not positioned and secured as specified for level A.
111. Packing not in accordance with the referenced document for industrial.
112. Marking missing, illegible, incorrect, or incomplete for level A or industrial.

113. Preservation not as specified.

Minor

204. Marking not as specified.

5. PACKAGING

5.1 Preservation (see 6.6). Components comprising a complete detecting set

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shall be placed within the case as provided for. The case shall be closed andsecured by the attached clasps. A minimum of two tamperproof seals shall be attached to the clasps in a manner that the case cannot be opened without destroying the seals.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Each detecting set shall be individually packed in a close-fitting box conforming to PPP-B-636, V3c. Closure shall be in accordance with the appendix to the box specification. Unless otherwise specified (see 6.2), the packed detecting sets shall be palletized. The pallets shall conform to NN-P-71, Type IV. Load arrangement shall conform to pallet pattern 3, Appendix D, of MIL-STD-147. The height of the pallet including the load shall not exceed 43 inches. The detecting sets shall be secured to the pallet with a total of five primary and secondary straps. Straps shall conform to QQ-S-781, Type I or IV, Finish A. Wood battens of not less than 1/2-inch thickness and not less than 2 inches wide shall be positioned under the vertical portions of the straps and straps held in place by staples.

5.2.2 Commercial. Detecting sets shall be packed in accordance with MIL-STD-1188.

5.3 Marking (see 6.6).

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. This specification covers a detecting set intended for use over all types of terrain for the location of buried metallic antipersonnel and antitank mines.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
c. Whether batteries are to be furnished (see 3.1).
d. Time frame required for submission of the preproduction model and number of detecting sets to be furnished (see 3.2).
e. Whether the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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f. Whether initial production inspection is required and number of detecting sets to be furnished (see 3.3).
g. Degree of packing required (see 5.2).
h. Whether palletization is required (see 5.2.1).

6.3 Definitions.

6.3.1 Nonmetallic medium. Air, all types of soils including those having a magnetic volume susceptibility of between 13,000 and 15,000 micro cgs units, fresh or sea water, ocean beaches, wood, and all surfacings except those with metal reinforcing.

6.3.2 Output voltage. The voltage developed across the head set.

6.3.3 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number, and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.4 Preproduction model. Any changes or deviations of production detecting sets from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish detecting sets conforming to this specification.

6.5 Government-furnished facilities. When requested by the contractor, the contracting officer should contact the U.S Army Mobility Equipment Research and Development Command, ATTN: DRDME-NE, Fort Belvoir, VA 22060, to determine the availability of the Magnetic Test Lane at Fort Belvoir, VA, for use by the contractor in performance of the test required in 4.6.2.12 (see 4.6.2).

6.6 Preservation. Degrees of preservation have not been included as the preservation specified is the minimum acceptable for any degree of preservation considered level A.

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each detecting set.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of detecting sets.

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6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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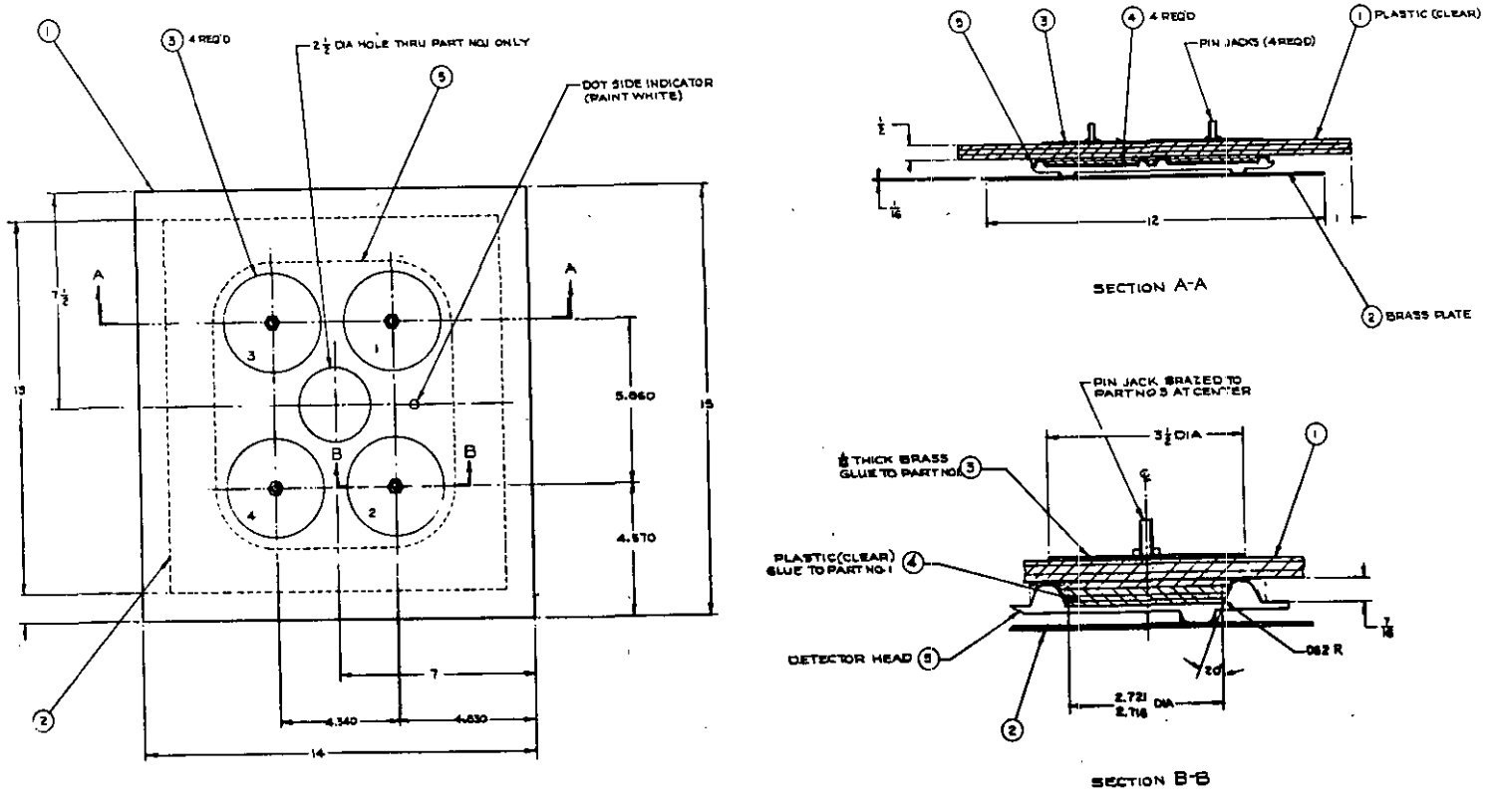


FIGURE 1. SEARCH HEAD TEST JIG (TRANSISTORIZED MINE DETECTOR)

NOTES:

1. Unless otherwise specified, all dimensions are in inches.
2. Unless otherwise specified, tolerance on inch-dimensions shall be ± 0.062.

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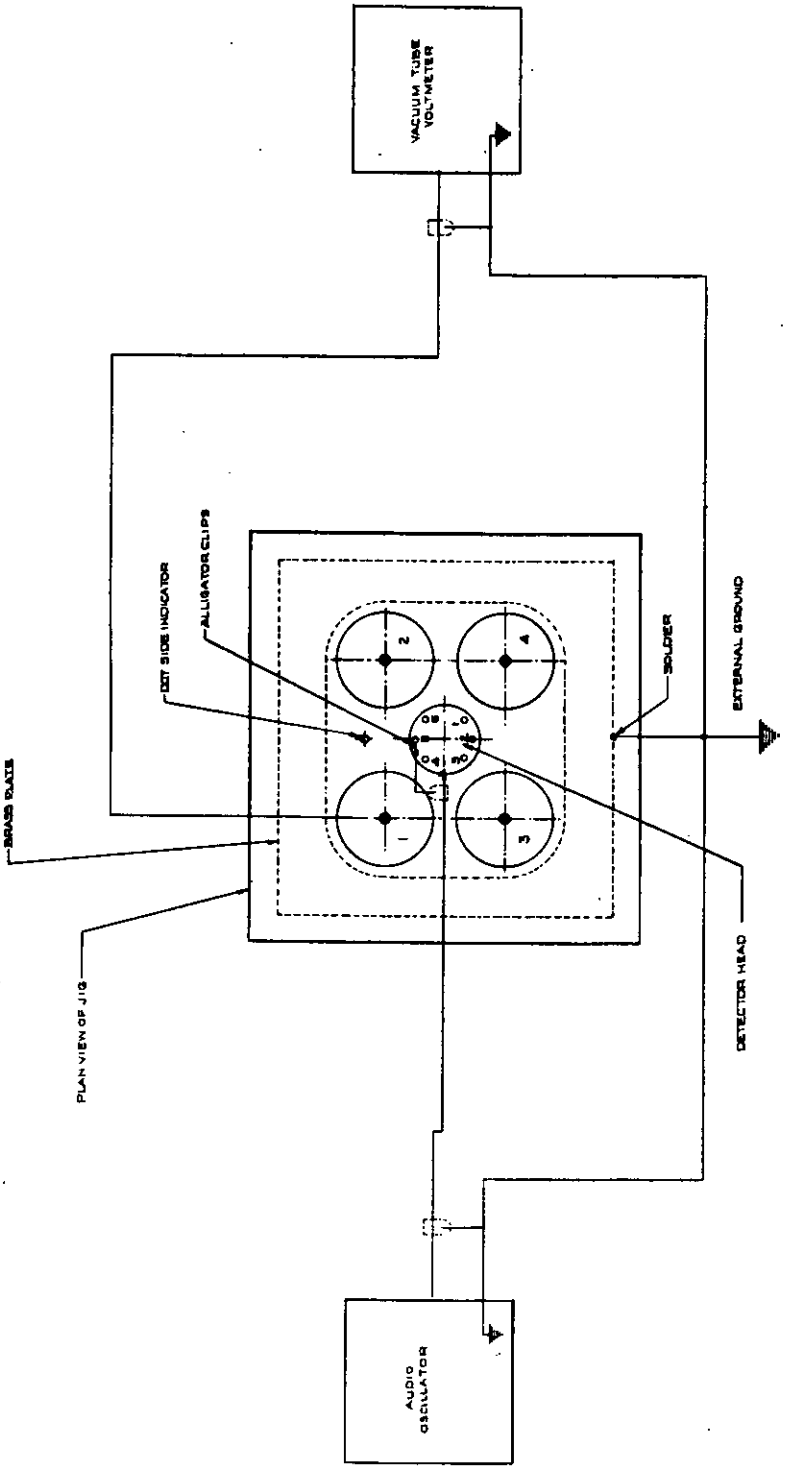


FIGURE 2. SEARCH HEAD TOROIDAL WINDING TEST CIRCUIT (TRANSISTORIZED MINE DETECTOR)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-0023359E(ME) Detecting Set, Mine, Portable, Metallic

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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